

LEAF, CONFIGURATION AND SYSTEM

Final RTL evidence | 2026-09-30 | Group 3/3: seven named benches

BENCH / OBSERVED LOG COUNTS	REFERENCE CHECK / COVERAGE BOUNDARY
tree_mapping_tb 12,904 stimulus rows Observed: TREE_MAPPING_COMPLETE checks=12904	Loop popcount and direct cell sums; T/G/R + invalid-ID comparison. This archive is RTL, not mapped-netlist proof.
review_leaf_tb 91,142 check calls Observed: REVIEW_LEAF_COMPLETE	Popcount/thermometer rules, address bounds, allocation wrap and floor oracle. The logged count is assertion calls.
review_dither_tb 7 amplitude/seed conditions accepted/condition: 196,866 or 200,000 Observed: REVIEW_DITHER_COMPLETE	Support/PMF/mean/lag-1 checks; hold and seed-reset invariants. Accepted draws differ from 200k clock cycles.
p1_tb 464,823 TB checks; 0 errors Observed: P1 RESULT: PASS	M1-M5 golden vectors plus structural properties, dither support and connected chain. 464,823 is the TB check counter, not codes.
p2_periph_tb 709 TB checks; 0 errors Observed: p2_periph PASS	Directed guards, sticky status, RDAC fanout, phase schedule and register rules. Full-table load cannot trigger unreachable sum guard.
p2_smoke_tb 1,154 TB checks; 0 errors Observed: p2_smoke PASS	Configuration readback; allocator invariants and selected popcount inputs. Smoke coverage, not complete acceptance.
p3_top_tb 16,386 TB checks; 0 errors 4,095 output rows; 16-cycle spacing Observed: P3 TOP RESULT: PASS	File oracle for dout/clip + cumulative OR for sticky analog-overflow status. Forced dither; file row 0 skipped; no dither-source test.

Counts retain their own units. No combined 'total coverage' score. All 21 log and TB hashes verified.

Source: docs/evidence/20260930/final_rtl/ | Zero-delay RTL logs, not STA or analog performance.